Polarization (Griffiths §4.1): structured notes

Griffiths, *Introduction to Electrodynamics* (5th ed.), pp. 166–172. Full content of the study notes, arranged by topic for revision. Equation numbers match the book. Figures are sketches, not the book's figures.

1. Dielectrics: why insulators respond to a field

- **Conductor:** some electrons are free to move through the whole body.
- **Insulator (dielectric):** every charge is bound to an atom or molecule.
- A bound charge cannot leave, but it can shift slightly within its atom or molecule. The cumulative effect of these small shifts gives the characteristic behaviour of dielectrics.
- Two mechanisms distort the charge distribution: *stretching* (induced dipoles, §2) and *rotating* (alignment of permanent dipoles, §3).

2. Induced dipoles

2.1 Physical picture

- A neutral atom has a nucleus of charge $+q$ and an electron cloud of total charge $-q$. Net charge is zero.
- In a field $\mathbf{E}$, the nucleus is pushed along $\mathbf{E}$ and the cloud is pushed against it. The two charge centres separate by a distance d.
- The separated charges attract each other and pull back. The atom reaches equilibrium when the pull from the cloud equals the push from the field.
- If the field is large enough, the atom is ionised (the electrons are pulled away). Below that, equilibrium is established.
- Result: the atom is polarised and carries a dipole $\mathbf{p} = q\mathbf{d}$, pointing along $\mathbf{E}$.

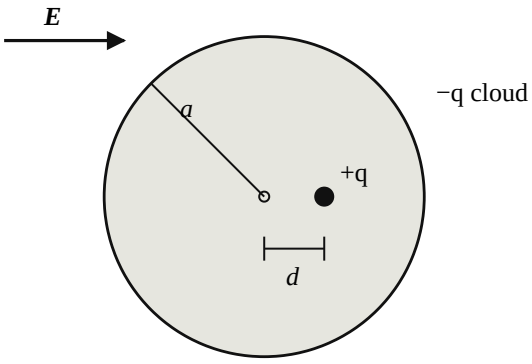


Figure 1. My sketch of the Example 4.1 model: the nucleus is pushed right by E and the cloud pulls it back.

2.2 Field of a uniformly charged ball (Gauss's law)

Model the electron cloud as a uniform ball of total charge $-q$ and radius a. Take a point at distance $s < a$ from the centre.

- Charge inside radius s: the ball is uniform, so this fraction is $(s/a)^3$. Enclosed charge $= q\,s^3/a^3$.
- Gauss's law over a sphere of radius s: $\oint \mathbf{E}\cdot d\mathbf{a} = Q_{\text{enc}}/\epsilon_0$.

$$E\cdot 4\pi s^2 = \frac{q\,s^3}{a^3\,\epsilon_0} \quad\Longrightarrow\quad E = \frac{1}{4\pi\epsilon_0}\frac{q\,s}{a^3}.$$

At $s = d$, the field grows linearly with d. The restoring pull is proportional to the displacement, as for a spring.

2.3 Equilibrium and polarizability (Eqs. 4.1, 4.2; Example 4.1)

- Equilibrium: applied field equals the cloud's field at the nucleus, $E = E_e$.
- Substitute E_e from §2.2 with $s = d$:

$$E = \frac{1}{4\pi\epsilon_0}\frac{q\,d}{a^3} \quad\Longrightarrow\quad p = qd = (4\pi\epsilon_0 a^3)\,E.$$

> **RESULT**
> $\mathbf{p} = \alpha\mathbf{E}$ (4.1), with $\alpha = 4\pi\epsilon_0 a^3 = 3\epsilon_0 v$ (4.2), where $v = \frac{4}{3}\pi a^3$ is the volume of the ball.

- α grows with the volume of the atom: larger atoms are easier to polarise.
- Accuracy: the uniform-ball model is crude. The book states it is accurate to within a factor of about 4 for many simple atoms.

2.4 Units of α

- p is in C·m and E is in N/C. So $\alpha = p/E$ is in C²·m/N.
- Table 4.1 gives $\alpha/4\pi\epsilon_0$ in $10^{-30}\,\mathrm{m}^3$. $4\pi\epsilon_0$ has units C²/(N·m²); multiplying by m³ gives C²·m/N. The two agree.

2.5 Molecules: polarizability depends on direction

- CO_2 is easier to stretch along its axis than across it: $\alpha_\parallel \approx 4.5\times10^{-40}$ and $\alpha_\perp \approx 2\times10^{-40}$ C²·m/N.
- Split $\mathbf{E}$ into parts parallel and perpendicular to the axis: $\mathbf{p} = \alpha_\parallel\mathbf{E}_\parallel + \alpha_\perp\mathbf{E}_\perp$. In general $\mathbf{p}$ need not point along $\mathbf{E}$.
- For an asymmetric molecule, each component of $\mathbf{p}$ depends on all components of $\mathbf{E}$. This needs nine constants α_{ij}, the polarizability tensor:

$$p_x = \alpha_{xx}E_x + \alpha_{xy}E_y + \alpha_{xz}E_z,$$

and similarly for p_y, p_z (4.3).

- Each row of the 3×3 array gives one component of $\mathbf{p}$. The off-diagonal terms vanish in principal axes.

2.6 Problem 4.2: the electron cloud is not a uniform ball

- Hydrogen ground-state cloud: $\rho(r) = \dfrac{q}{\pi a^3} e^{-2r/a}$.
- Method: find $E_e(r)$ with Gauss's law using the shell-by-shell enclosed charge, then expand the exponential for $r \ll a$: $e^{-2r/a} \approx 1 - 2r/a$.
- The field is again linear in the displacement, so $\mathbf{p} \propto \mathbf{E}$ still holds in the weak-field limit. Only the constant changes.

3. Permanent dipoles

3.1 Physical picture

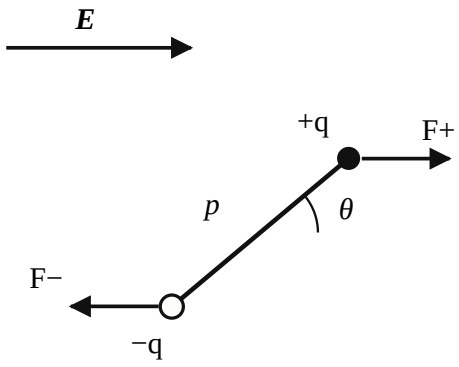
- Water is bent at about $105°$. Oxygen draws electrons toward itself, so one end is negative and the other positive.
- Water therefore has a permanent dipole $\mathbf{p}$ even without a field. Its dipole moment is 6.1×10^{-30} C·m.
- This dipole is why water dissolves salts well.

3.2 Torque (Eq. 4.4)

- Uniform field: force on $+q$ is $q\mathbf{E}$; force on $-q$ is $-q\mathbf{E}$. Net force is zero.
- The forces act at different points, so they produce a torque. Take the centre as origin; the ends are at $\pm\mathbf{d}/2$.

$$\mathbf{N} = \frac{\mathbf{d}}{2} \times q\mathbf{E} + \left(-\frac{\mathbf{d}}{2}\right) \times (-q\mathbf{E}) = q\,\mathbf{d} \times \mathbf{E} = \mathbf{p} \times \mathbf{E} \quad (4.4).$$

- The two contributions point the same way and add.
- The torque turns $\mathbf{p}$ toward $\mathbf{E}$. Magnitude: $pE\sin\theta$, where θ is the angle between $\mathbf{p}$ and $\mathbf{E}$.



Net force zero; torque turns p toward E

Figure 2. My sketch of a permanent dipole in a uniform field. The two forces are equal and opposite, but they act at different points, so they turn the dipole.

3.3 Potential energy (Eq. 4.6)

- To turn the dipole slowly from $90°$ to θ, an external agent must supply a torque $pE\sin\phi$ against the field.

$$W = \int_{\pi/2}^{\theta} pE\sin\phi\, d\phi = -pE\cos\theta = -\mathbf{p}\cdot\mathbf{E}.$$

> **RESULT**
> $U = -\mathbf{p}\cdot\mathbf{E}$ (4.6), with the zero of energy at $\theta = 90°$. The lowest energy is at $\theta = 0$, so a free dipole aligns with $\mathbf{E}$.

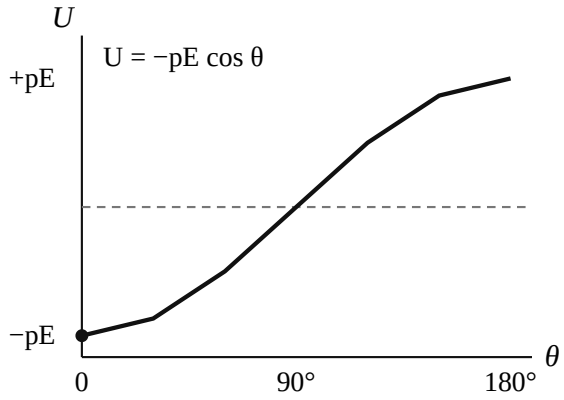


Figure 3. My sketch of U against θ, with zero energy at 90°. The dipole is most stable at θ = 0, where U is lowest.

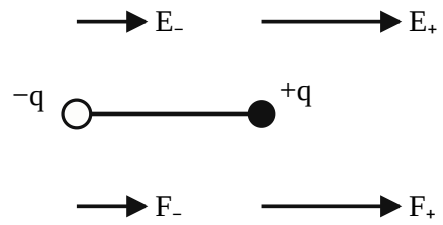
- This form applies to a permanent dipole of fixed size (book footnote 3).
- An induced dipole depends on E, so $-\mathbf{p}\cdot\mathbf{E}$ is not the full energy for it.

3.4 Non-uniform field: a net force (Eq. 4.5)

- Let the field at the positive end be $\mathbf{E}(\mathbf{r}+\mathbf{d}/2)$ and at the negative end $\mathbf{E}(\mathbf{r}-\mathbf{d}/2)$.
- For a short dipole, the difference is a first-order change along $\mathbf{d}$: $\Delta\mathbf{E} \approx (\mathbf{d}\cdot\nabla)\mathbf{E}$.
- Force: $\mathbf{F} = q\,\Delta\mathbf{E}$.

$$\mathbf{F} = (\mathbf{p}\cdot\nabla)\mathbf{E} \quad (4.5).$$

- $\nabla = \hat{x}\,\partial_x + \hat{y}\,\partial_y + \hat{z}\,\partial_z$ is the vector of partial derivatives. Partial derivatives are not part of Class 12 maths and must be learned first.
- The force is significant only if the field changes appreciably over one molecule, which is rare in dielectrics.



$E_+ > E_-$, so $F_+ > F_-$: net force to the right

Figure 4. My sketch of a short dipole in a non-uniform field. Each end feels its own field, and the stronger field at the + end gives the bigger force.

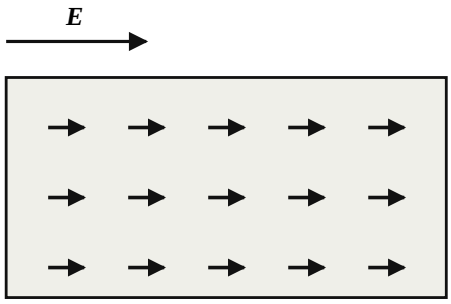
Book footnote 2 notes that $\nabla(\mathbf{p}\cdot\mathbf{E})$ would be shorter, but $(\mathbf{p}\cdot\nabla)\mathbf{E}$ is the safe form when $\mathbf{p}$ varies with position.

4. Polarization $\mathbf{P}$

- Definition: $\mathbf{P}$ = dipole moment per unit volume. Unit: C/m².
- If N identical dipoles $\mathbf{p}$ lie in unit volume and point the same way:

$$\mathbf{P} = N\mathbf{p}.$$

- Two mechanisms give a net $\mathbf{P}$ along $\mathbf{E}$: induced dipoles point along $\mathbf{E}$ (Eq. 4.1); permanent dipoles turn toward $\mathbf{E}$ (Eq. 4.4).



P = N p (dipoles per volume × dipole each)
all dipoles point along E; the arrows are the dipoles

Figure 5. My sketch of aligned dipoles in a block. Polarization is the total dipole moment per unit volume.

- The field produced by the material depends on $\mathbf{P}$, not on each atom separately. This is the link to §4.2 of the book.
- Some materials keep $\mathbf{P}$ after the field is removed (book, in passing). This is outside the first module.

4.1 Why alignment is never complete

- Energy of a permanent dipole in a field: $-pE\cos\theta$.
- Thermal energy per molecule: of order k_BT, about 0.026 eV at room temperature.

- If $pE \ll k_B T$, thermal motion dominates and the dipoles point in all directions. For ordinary fields pE is small, so the net alignment is weak.
- The book notes that alignment is lost almost at once when the field is removed.

5. Open points (to verify before exam use)

- The "factor of about 4" accuracy of Eq. 4.2 is quoted from the book's statement. A possible reason is that the real cloud is concentrated nearer the nucleus than a uniform ball. This is not verified.
- Eq. 4.5 requires partial derivatives and ∇. Revise these before attempting the derivation from first principles.
- Problem 4.2 is set out but the integral is not carried through in these notes.